Tht Yuo Cna Reda Ths Snetnce
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NP-Cmoplet Prbolems

Optimal Measurements for the Dihedral Hidden Subgroup Problem

Dave Bacon, Santa Fe Institute

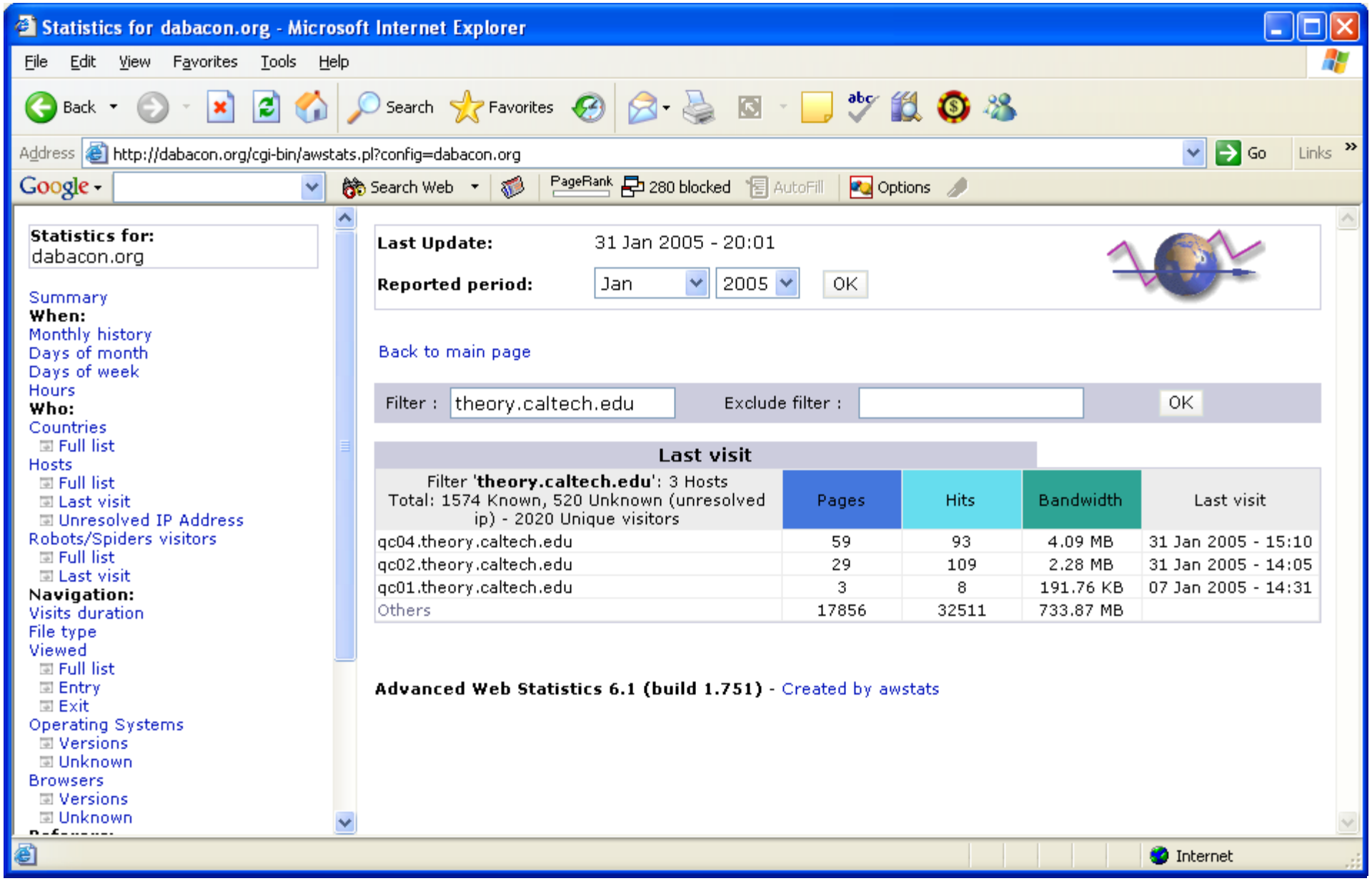


in collaboration with
Andrew Childs, Caltech
Wim van Dam, U.C. Santa Barbara



SQuint

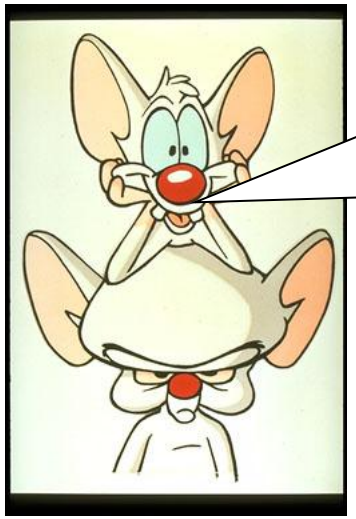
Southwest Distractions



Quantum Algorithms



"If I had a good idea in quantum algorithms that's what I'd be working on" – John Preskill, 2004



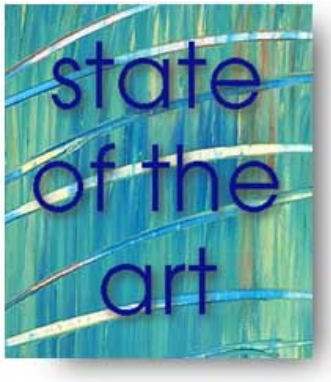
What happens to quantum computing if no significant new algorithms are discovered?

When I was a Grad Student…

Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]

State of the Art



Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …

The Problem

The Hidden Subgroup Problem (HSP)

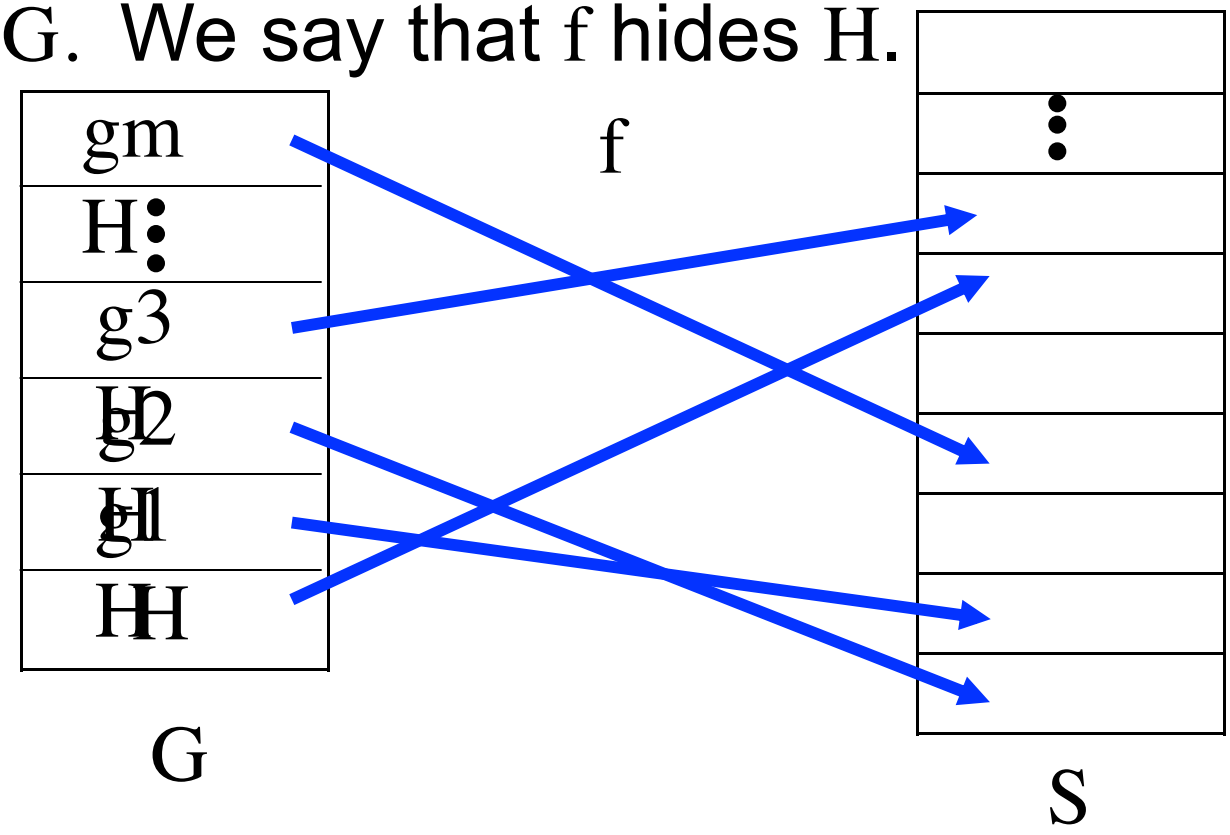
Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H.

Goal: Find (a generating set for) H.

Efficient algorithm: run time $\text{poly}(\log |G|)$.

Anatomy of Our Problem

Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H.



The Hidden Subgroup Saga

Why is everyone so interested in the HSP?

1. Factoring can be solved by solving an HSP for an Abelian group.
2. Solving the HSP for the symmetric group would given an algorithm for graph isomorphism.
3. Solving the HSP for the dihedral group would give algorithms for k-unique shortest vector in a lattice problems (and thus break a cryptosystem.)
4. Solving the HSP would make everyone work on really new quantum algorithms

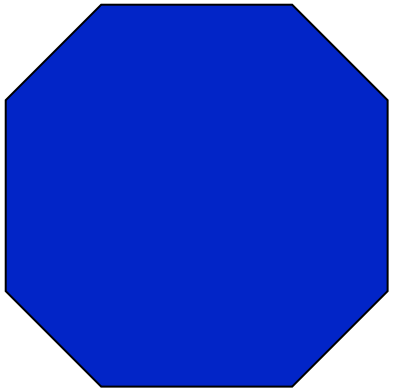
The Hidden Subgroup Success

- Abelian groups [Shor 94; Kitaev 95]
- $\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$ [Püschel, Rötteler, Beth 98]
- Normal subgroups [Hallgren, Russell, Ta-Shma 00]
- "Amost Abelian" groups [GSVV 01]
- $\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$ [FIMMS 02]
- "Near Hamiltonian" groups [Gavinsky 04]
- Some q-hedral groups [MRRS 04]
- Som $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$ [Inui, Le Gall 04]
  e

Things are heating up

A Strange Lock





Rotation : | 0 | 0 | 0 |

Reflection: | 1 |

Display: | BFD |

Setup: Display will give same result if we enact rotation by k and if we enact rotation by -k+x and a reflection.

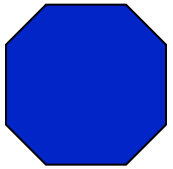
x is the hidden combo.

Goal: Make display repeat to open the lock.

Dihedral Hidden Reflection Problem

Dihedral HSP

Dihedral group



D8

G = DN: symmetry group of a N-sided regular polygon.

Rotation ≡ s, sN=e

Reflection ≡ r, r2=e and rsr=s-1

Element of DN: rtsk, where t∈□2,s∈□N

Theorem: Solving the hidden subgroup problem when H={e,rsl}, l∈□N or H={e} solves the full Dihedral HSP [Etinger, Hoyer 98]

Dihedral HSP Facts

Efficient algorithm for DHSP

[Regev 02]

Efficient algorithm for the poly(d) unique shortest vector in a lattice problem

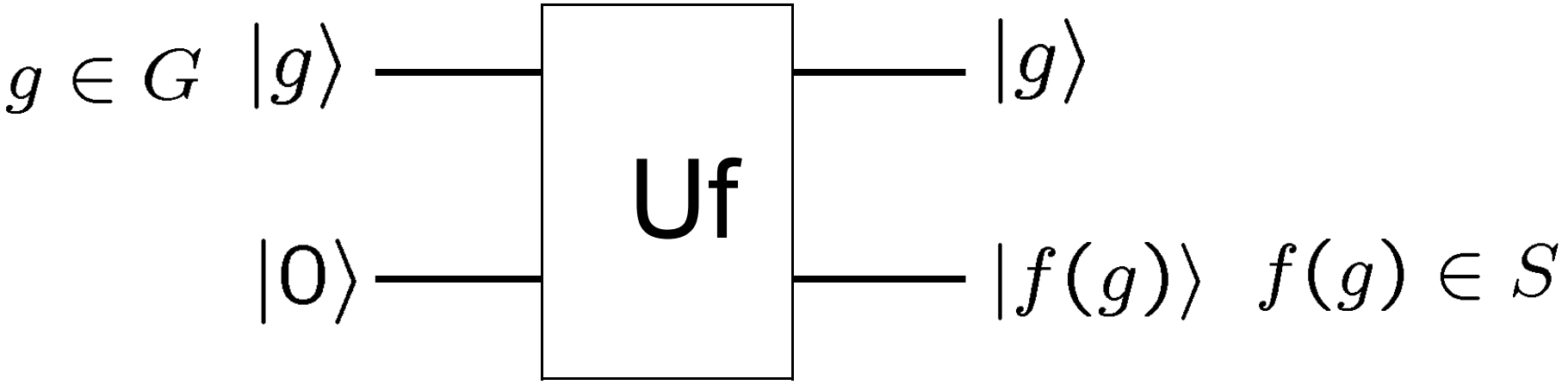
Quantum algorithm: $2^{O(\sqrt{N})}$ time [Kuperberg 03]
poly(log(N)) space [Regev 04]

Efficient algorithm average case subset sum

[Regev

Efficient quantum Algorithm for DHSP

Standard Query Model



$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|G|}} |g\rangle|f(g)\rangle$$

constant and distinct on different left cosets of H:

$$\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH| \qquad |gH\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |gh\rangle$$

hidden subgroup state

HSP as State Distinction

Single Copy

Given: $\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH|$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Multiple Copy

Given: $\rho_H^{\otimes k}$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Query Complexity of HSP

Given: *k=O(log(N))* standard queries to HSP

Exists: an inefficient quantum algorithm which

solves any HSP. [Ettinger, Hoyer, Knill 99]

"Query complexity of HSP is polynomial."

Results: Shor's algorithm is optimal with O(1) queries to the HSP [Ip 03]

Question: Are there groups for which *O(log(N))* standard queries are necessary?

Holevo's Condition

Given: ρ_i with probability p_i

Measure: POVM with elements $\{G_j\}$

Outcome: j corresponds to guessing state ρ_j

Maximize: probability of successfully identifying ρ_j

Necessary and Sufficient Condition: [Holevo:73]

$$\left(\sum_i p_i \rho_i G_i - p_j \rho_j\right) G_j = G_j \left(\sum_i p_i \rho_i G_i - p_j \rho_j\right) = 0$$

- **Hard to find explicit forms for *Gj***

Pretty Good

Given: ρ_i with equal probabilities

Observation: the "Pretty Good Measurement" is often pretty good.

Pretty Good Measurement (PGM):

$$G_j = T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i \rho_i$$

(plus $I - \sum_j G_j$ if needed.)

Hidden Reflection

Given: $\rho_H^{\otimes k}$ with equal a priori probability

$$H = \{e, rs^l\}, l \in \mathbb{Z}_N$$

Theorem: [Bacon, Childs, van Dam 05]

The PGM is the optimal measurement for identifying *H*.

Theorem: [Moore, Russell 05]

The PGM is the optimal measurement for identifying *H* when *H* and *G* are a Gelfand pair.

arXiv generalization speed = Jan 10 – Jan 31 = 21 days

One Copy

Punishment

One copy of the Hidden Subgroup state:

$$\rho_d = \frac{1}{N} \sum_{k \in \mathbb{Z}_N} |\phi_{d,k}\rangle\langle\phi_{d,k}|$$

$$|\phi_{d,k}\rangle = \frac{1}{\sqrt{2}} (|k, 0\rangle + |-k + d, 1\rangle)$$

Basis char

$$\rho_d = \frac{1}{N} \sum_{k \in \mathbb{Z}_N} |\tilde{\phi}_{d,k}\rangle\langle\tilde{\phi}_{d,k}| \qquad \omega = \exp\left(\frac{2\pi i}{N}\right)$$

$$|\tilde{\phi}_{d,x}\rangle = |x\rangle \frac{1}{\sqrt{2}} \left(|0\rangle + \omega^{xd}|1\rangle\right)$$

$$\rho_d = \frac{1}{2N} \sum_{x \in \mathbb{Z}_N, b, c \in \mathbb{Z}_2} \omega^{(b-c)xd} |x, b\rangle\langle x, c|$$

Multi Copy

More Punishment

$$\rho_d^{\otimes k} = \frac{1}{(2N)^k} \sum_{x \in \mathbb{Z}_N^k} \sum_{b,c \in \mathbb{Z}_2^k} \omega^{((b-c)\cdot x)d} |x, b\rangle\langle x, c|$$

$$\rho_d^{\otimes k} = \frac{1}{(2N)^k} \sum_{x \in \mathbb{Z}_N^k} \sum_{p,q \in \mathbb{Z}_N} \sqrt{\eta_p^x \eta_q^x}\, \omega^{(p-q)d} |S_p^x\rangle\langle S_q^x|$$

$$|S_t^x\rangle := \frac{1}{\sqrt{\eta_t^x}} \sum_{b \in S_t^x} |x, b\rangle \quad S_t^x := \{b \in \mathbb{Z}_2^n | b \cdot x = t\}$$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

Optimal PGM

Stop The Punishment

$$E_j = \frac{1}{N} \sum_{x \in \mathbb{Z}_N^k} \sum_{p,q \in \mathbb{Z}_N} \omega^{(p-q)j} |S_p^x\rangle\langle S_q^x|$$

$$|S_t^x\rangle := \frac{1}{\sqrt{\eta_t^x}} \sum_{b \in S_t^x} |x, b\rangle$$

$$S_t^x := \{b \in \mathbb{Z}_2^n | b \cdot x = t\}$$

Note: Entangled measurement over k copies

Success Probability

Probability of correctly guessing reflection:

$$p = \frac{1}{2^k N^{k+1}} \sum_{x \in \mathbb{Z}_N^k} \left(\sum_{t \in \mathbb{Z}_N} \sqrt{\eta_t^x} \right)^2$$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

p is related to a phase transition

Subset Sum Problem

Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$ i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Goal: Find a subset of *x* which sums to *t*

$$\sum_{x_i \in S} x_i = t \mod N$$

Problem: does there exists such a subset?

NP-complete

Problem: If such a subset exists, return this subset

NP-hard

basis for old (broken) public key cryptosystem

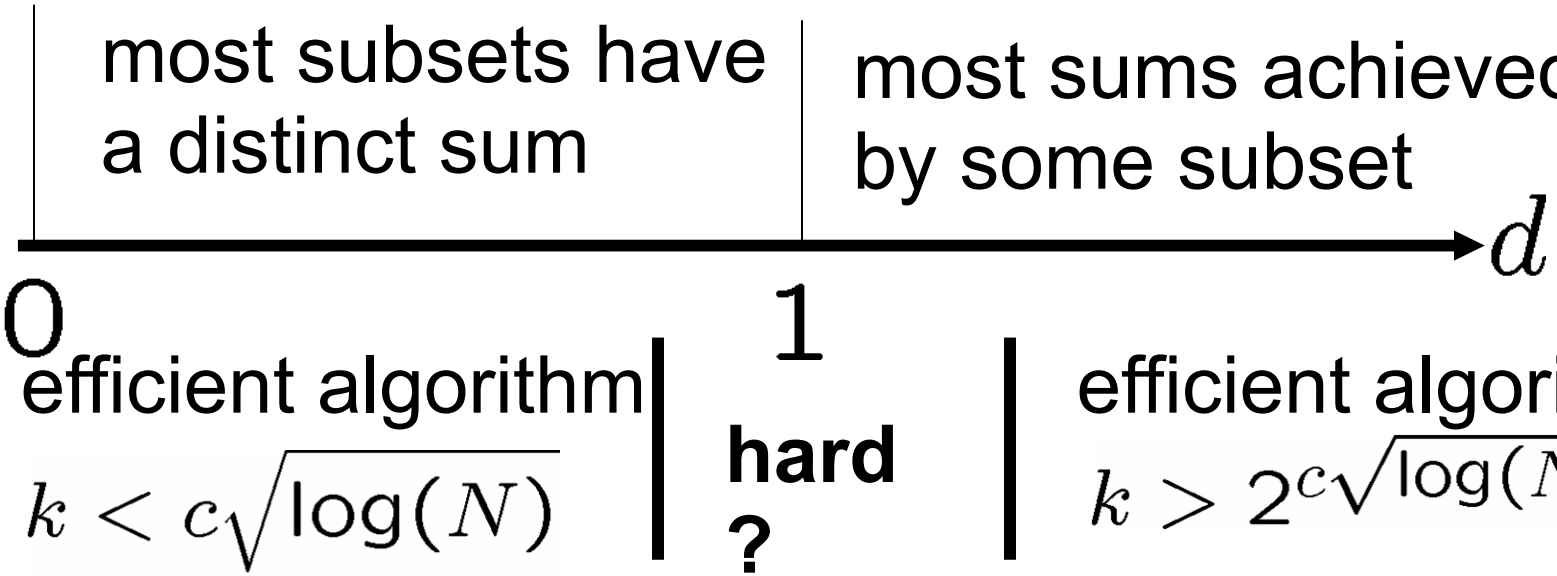
The Subset Sum Landscape

Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$

i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Define: density := $d = \frac{k}{\log N}$

"low density" **"high density"**



[Lagarias, Odlyzko

[Flaxman, Pryzdatek 04]

Success Probability

Probability of correctly guessing reflection:

$$p = \frac{1}{2^k N^{k+1}} \sum_{x \in \mathbb{Z}_N^k} \left(\sum_{t \in \mathbb{Z}_N} \sqrt{\eta_t^x} \right)^2$$

$\mathbb{Z}_N^k :=$ k dimensional vector of elements of $\mathbb{Z}_N$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

Two orders:

For $k<N$, p is exponentially small (in log(N))

For $k>N$, p is $O(1)$

transition at density = *d=1*

Must Work Hard

Theorem: [Bacon, Childs, van Dam 05]
Determining the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

Theorem: [Bacon, Childs, van Dam 05]
Determining the most significant bit of the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

An efficient quantum circuit for the dihedral PGM?

MMMMore



- Implementation of dihedral PGM in a particular manner would allow for solution of subset sum problems.
- Information theoretic description of bounds on number of copies to solve dihedral HSP.

Conclusions

- Optimal measurements for the dihedral HSP
- Probability of success and subset sum
- At k=log(N) copies transition to solvable

Open

- Efficient quantum circuit for optimal measurement
- Efficient quantum circuits for a PGM

Optimal Measurements for the Dihedral Hidden Subgroup Problem

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SQuint